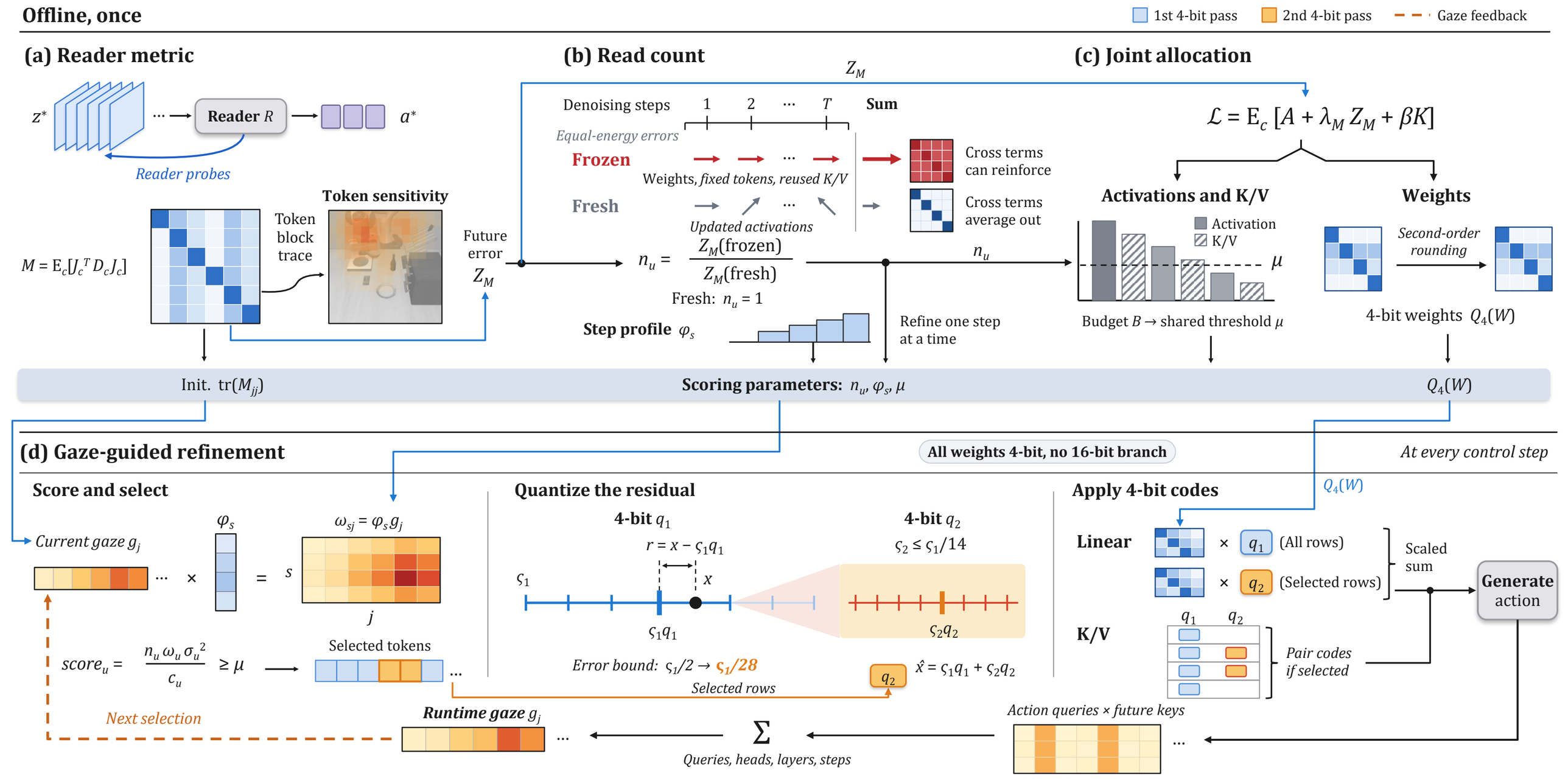

Offline, once
1st 4-bit pass
2nd 4-bit pass
Gaze feedback
(a) Reader metric
Reader R
Reader probes
Token sensitivity
Token block trace
Future error
(b) Read count
Denoising steps
Sum
Equal-energy errors
Frozen
Weights, fixed tokens, reused K/V
Fresh
Updated activations
Cross terms can reinforce
Cross terms average out
Fresh: n_u = 1
Step profile φ_s
Refine one step at a time
(c) Joint allocation
Activations and K/V
Weights
Activation
K/V
Budget B → shared threshold μ
Second-order rounding
4-bit weights Q_4(W)
Init. tr(M_jj)
Scoring parameters: n_u, φ_s, μ
(d) Gaze-guided refinement
All weights 4-bit, no 16-bit branch
At every control step
Score and select
Current gaze g_j
Selected tokens
Next selection
Quantize the residual
4-bit q_1
4-bit q_2
Error bound: ς_1/2 → ς_1/28
Selected rows
Apply 4-bit codes
Linear
(All rows)
(Selected rows)
Scaled sum
K/V
Pair codes if selected
Generate action
Runtime gaze g_j
Queries, heads, layers, steps
Action queries × future keys